

Cultural Matching and the Persistence of Social Ties

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Abstract

How ~~do cultural tastes~~ does cultural matching shape the evolutionary dynamics of personal networks during major life transitions? While ~~traditional perspectives often view tastes primarily as malleable byproducts of network transmission, we~~ network theories frequently emphasize the structural and demographic drivers of tie dynamics, I advance an integrative ~~cultural-matching~~ framework in which ~~pre-existing tastes and cultural practices actively guide tie retention~~ cultural matching actively protects personal ties from decay alongside structural opportunity. Using longitudinal ego-network data from the *NetSense* study, which tracked college students across eight survey waves, ~~we find that shared cultural tastes—both broad categorical interests and specific shared activities—significantly reduce~~ I find that cultural matching—both in broad categorical domains and in specific leisure activities—significantly reduces the hazard of tie ~~dissolution~~ decay, net of dyadic structural embeddedness (triadic closure), residential proximity, and demographic homophily. Conversely, ~~cultural network opacity, or a lack of mutual cultural knowledge~~ I find that cultural network opacity, or ego’s lack of knowledge of the alter’s tastes, acts as an independent relational liability that accelerates tie decay. Crucially, ~~these stabilizing effects are the~~ protective effect from tie decay via cultural matching is heavily concentrated among subjectively close ties: cultural ~~alignment-matching~~ forms an essential interactional core for maintaining ~~core~~ close relationships amidst high background churn, whereas weak ties face such a high baseline rate of decay that cultural matching alone ~~cannot~~ does little to sustain them. These findings ~~demonstrate how~~ show that, within high-turnover peer-network ecologies, individuals draw upon cultural ~~dispositions as portable resources~~ matching as a portable resource to selectively curate and ~~stabilize~~ protect their most meaningful social bonds from decay.

1 Introduction

A central issue in the study of culture and networks concerns the co-evolution of social ties and cultural tastes (Lizardo, 2006; Edelman and Vaisey, 2014). ~~Sociological perspectives on taste diffusion often posit that an individual’s cultural profile is continuously molded by their current social network configuration (Mark, 1998b; McPherson, 2004). In this view, tastes diffuse across existing homophilous ties, which are themselves structured by socio-demographic similarity (Lazarsfeld and Merton, 1954; McPherson et al., 2001; Feld and Grofman, 2001). Under this network-transmission imagery, tastes are often construed as fluid, network-contingent contents liable to shift as local network configurations evolve (Mark, 2003). Yet longitudinal research reveals~~ Contemporary research in network science and computational social science has shown that relational structures and cultural practices evolve via dynamic, reciprocal feedback (Mark, 1998b; McPherson, 2004; Lewis and Kaufman, 2018). Social ties provide the structural channels and interactional opportunities through which cultural information and practices diffuse and are subsequently adopted (Lazarsfeld and Merton, 1954; Carley, 1991; Borgatti, 2001), and cultural profiles guide partner selection and tie formation, generating choice and status homophily thus reinforcing network boundaries (DiMaggio, 1987; McPherson et al., 2001; Shalizi and Thomas, 2007).

Yet a long-standing finding in ego-network research is that personal networks are frequently in flux characterized by pervasive turnover, with ties constantly being formed, renegotiated, and dissolved—particularly pruned—particularly during critical life-course transitions such as entry-matriculating into college, job changes, or relocation getting married, having children, changing jobs, relocating, or retiring from paid work (Burt, 2000; Wellman et al., 1997; Kleinbaum, 2018; Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024). This pervasive network turnover raises an important substantive question: if personal networks undergo substantial turbulence during formative transitions, what explains the relative durability of individuals’ cultural dispositions, and how do individuals actively deploy these cultural profiles to navigate volatile social ecologies (Bourdieu, 1984; Holt, 1998)? endemic personal network volatility raises a key question: amid high background churn, how do individuals selectively decide which ties to preserve and which to allow to decay, and what role do cultural matching (Vaisey and Lizardo, 2010), structural embeddedness (Burt, 2000), and institutional foci (Feld and Grofman, 2009) play in protecting personal ties from decay?

In this paper, we evaluate I address this question by evaluating the empirical implications of a an integrative cultural matching model of tie persistence-decay within the dynamic setting of an undergraduate-cohorta sample of students during the transition to college. Rather than treating tastes solely as passive by-products of existing networks, we propose that pre-existing cultural dispositions serve as an active micro-interactional “social glue” that helps determine which network ties survive and which dissolve amidst high-baseline churn (Burt, 2000). cultural tastes as passive byproducts of network positions, treating network ties as static backcloths that only transmit information but don’t themselves change, or framing culture and structure as competing explanations (Mark, 1998b), I examine how cultural matching operates in tandem with structural opportunity structures to modulate the dynamics of tie durability (Burt, 2000; Lewis and Kaufman, 2018; Rivera et al., 2010).

Under this framework, dyadic cultural matching (commonalities in tastes and leisure

practices) increases the over-time stability and serves as a vital micro-interactional anchor, increasing protection from tie decay, reducing interactional friction, generating positive emotional energy in shared conversational rituals (Collins, 2004; Lizardo, 2016), elevating the cognitive salience of social ties, operating in tandem with structural and dyadic factors such as subjective closeness, spatial propinquity, and tie duration. Using alters, and ultimately rendering relationships, particularly the most affectively meaningful ones, resilient against background turnover (DiMaggio, 1987; Lizardo, 2006; Meuleman, 2021; Meuleman and Jæger, 2022). Crucially, I examine cultural matching not in a structural vacuum, but alongside key dimensions of positional and collective embeddedness—including triadic closure (shared contacts in the ego network), spatial foci (residential dorm co-presence), subjective closeness, and sociodemographic homophily.

1.1 Empirical Context

I use rich longitudinal ego-network panel data from the *NetSense* study over across eight survey waves (Bahulkar et al., 2016, 2017; Hachen et al., 2022), we track the tracking the decay and persistence of non-kin, on-campus peer ties across multiple college-year transitions. This empirical context provides a strategic window into relationship dynamics during emerging adulthood, capturing both the rapid reorganization of peer networks during the freshman transition and the subsequent consolidation of personal networks communities as graduation approaches. The *NetSense* design uniquely combines repeated ego-network nominations with granular measures of both broad categorical tastes and concrete, open-ended leisure activities, as well as complete alter-alter acquaintance networks that allow us to account for the structural embeddedness of each tie.

Empirically, we I examine the cultural matching mechanism along two three interrelated dimensions. First, we I establish the baseline protective effect of aggregate cultural matching—spanning cultural matching—evaluating both broad categorical domains domain matches and concrete leisure activities—on tie persistence activity matches—against tie decay, adjusting for dyadic structural embeddedness and residential proximity. Second, we I introduce the concept of “cultural network opacity” cultural network opacity—the degree to which an ego lacks mutual cultural knowledge about ego lacks knowledge of an alter’s preferences (indicated by substantive “Don’t Know” reports)—as preferences—as an independent predictor of tie decay. By evaluating Third, I investigate whether the protective capacity of cultural matching is concentrated in strong, subjectively close relationships or serves as compensatory scaffolding for weak ties to overcome what (Burt, 2000) calls the *liability of newness*. By testing these mechanisms in a discrete-time event history framework, we illuminate how event-history framework with ego fixed effects and cross-classified random effects models, I provide robust evidence that cultural matching operates as an active selection and retention mechanism within mechanism that protects ties from decay in dynamic peer environments.

The remainder of this article proceeds as follows. First, we outline the I develop an integrative theoretical framework linking tastes cultural matching, structural position, and tie persistence decay during life transitions. Next, we I describe the *NetSense* data and our panel data, measures of taste stability, and the discrete-time event history modeling strategy. Finally, we I present the empirical results and discuss their theoretical implications for the study of research on homophily, network turnover, and cultural capital.

2 Theoretical Framework

2.1 ~~Tastes~~ An Integrative Perspective on Culture and Social Networks: Two Perspectives Network Dynamics

The relationship between cultural tastes and social networks has ~~been conceptualized in two contrasting ways.~~ The *network transmission perspective*, drawing on constructural imagery (Carley, 1991), views tastes as malleable long been a central question at the intersection of cultural sociology and structural network analysis. Early formulations often treated culture and structure as competing explanations for social association (Erickson, 1996; Mark, 1998a, 2003; Lizardo, 2006). On one side, structural perspectives emphasize how exogenous opportunity structures, physical foci of activity (Feld, 1981), and spatial propinquity dictate interaction opportunities, with tastes diffusing secondarily across homophilous ties as fluid, network-contingent contents (Erickson, 1988, 1996; Mark, 1998a, 2003). In this view, social ties constitute the pre-existing “infrastructure” or “pipes” through which cultural practices diffuse (Borgatti, 2005; Erickson, 1996). Tastes are learned, adopted, and abandoned based on continuous exchange of cultural information and local bandwagon imitation within an actor’s immediate ego network (Mark, 2003). Because personal networks exhibit substantial over-time volatility—with between one and two-thirds of ties turning over annually in most personal networks (Morgan et al., 1997; Suitor et al., 1997)—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the *cultural capital perspective* conceptualizes tastes as highly (Carley, 1991; Borgatti, 2005). On the other side, cultural capital perspectives conceptualize tastes as durable, embodied dispositions acquired via early socialization (Bourdieu, 1984, 1990; Holt, 1998; Childress and Nault, 2019). Emphasizing the “interconvertibility” of different forms of capital (Bourdieu, 1986), this view sees cultural aptitudes not merely as by-products of ties, but as fundamental resources that help people form and sustain network connections. Here, networks are seen as fleeting structures built and maintained via interactions (DiMaggio, 1987). Consequently, individuals actively use pre-existing tastes to construct novel relationships or shed culturally incompatible ties, a dynamic formalized in the cultural matching model (Lizardo, 2006; Vaisey and Lizardo, 2010). In this framework, internalized cultural practices exert an independent causal effect on the formation of social networks because people self-select into ties based on the match between their tastes and those of others (Shalizi and Thomas, 2011). through early socialization that individuals actively deploy to select, maintain, and discard social ties (Bourdieu, 1984, 1990; Holt, 1998; Childress and Nault, 2019)

Building on this insight, recent theoretical elaborations specify how matching and influence operate simultaneously as dual mechanisms. For instance, Lewis and Kaufman (2018), building on early work by Lizardo (2006) propose a Recent advances in dynamic network modeling and digital trace research suggest that moving beyond a zero-sum opposition between culture and structure provides a more productive theoretical orientation (Lewis and Kaufman, 2018; Shalizi and Thomas, 2011; Edelman and Vaisey, 2014; Stark et al., 2020). Relational dynamics are fundamentally co-evolutionary: structural opportunity structures bring individuals into contact, while cultural matching determines whether those interactions

generate mutual affinity, emotional energy, and ongoing commitment (Collins, 2004; Lizardo, 2006). As Lewis and Kaufman (2018) propose in their generalized conversion model ~~showing that~~, cultural conversion (~~general connectivity as a result of having a wide variety of connectivity resulting from diverse ties~~) and cultural matching (forming ~~specific ties with others based on shared and sustaining specific ties based on convergent~~ tastes) operate ~~in tandem, constrained simultaneously, bounded~~ by the local cultural ecology (~~the distribution of particular tastes and cultural practices in the relevant setting~~). Crucially in our present study, the degree of cultural match at the dyadic level fundamentally determines whether relationships endure over time or instead selectively decay (Martin and Yeung, 2006). If tastes are durable and personal networks are volatile, it is highly likely that tastes serve as *drivers* of network connections rather than being determined exclusively by them. This leads to the concept of *choice homophily* or cultural matching: new social connections are selected and existing connections are maintained based on pre-existing cultural compatibility (Lazarsfeld and Merton, 1954; Werner and Parmelee, 1979). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact selection keyed to cultural alignment, rather than as an exogenous structural constraint (Carley, 1991; DiMaggio, 1987) and institutional ecology.

This co-evolutionary perspective is especially relevant when analyzing network evolution during major life transitions. Longitudinal studies consistently show that personal networks experience high turnover, with between one-third and two-thirds of active ties dissolving over relatively short intervals (Morgan et al., 1997; Suitor et al., 1997; Roy et al., 2022; Weiss et al., 2022). If social ties were the sole determinants of cultural tastes, individual cultural profiles would exhibit turbulence matching the rapid churn of their local networks. Yet empirical research indicates that core cultural orientations and aesthetic dispositions remain relatively stable across emerging adulthood (Vaisey and Lizardo, 2010; Childress and Nault, 2019). If personal networks are volatile while cultural dispositions are durable, cultural matching must serve as a critical sorting mechanism through which individuals actively navigate relational churn, selectively protecting culturally compatible ties while allowing mismatched or opaque connections to decay (DiMaggio, 1987; Lizardo, 2006).

2.2 Cultural Matching, Interaction Rituals, and Tie Persistence Decay

A ~~major challenge for any fundamental challenge confronting~~ newly formed social ~~relationship relationships~~ is what Burt (2000), drawing on Stinchcombe (1965), ~~refers to as terms~~ the “*liability of newness*.” ~~Newly formed Nascent~~ dyadic ties are fragile ~~and exhibit a high~~, characterized by ambiguous expectations, low emotional investment, and an elevated baseline hazard of decay (“*tie decay*”), ~~which steeply declines as a relationship matures and ripens into close friendship~~. What stabilizes a tie and helps it ~~As relationships mature, this decay hazard typically declines, but what enables a tie to overcome this liability of newness ?~~ While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we in the first place? While structural network theories highlight network constraint, multiplexity, and third-party monitoring (Coleman, 1988; Burt, 2000), I propose that *cultural matching* is a vital stabilizing mechanism. Rather than merely indicating demographic similarity, shared tastes and cultural practices serve as an active element that helps people establish basic

~~commonalities in outlook.~~ provides an indispensable micro-interactional foundation that protects ties from decay.

At the micro-level, ~~this matching stabilizes ties by reducing interactional friction, generating positive emotional energy during shared rituals (Collins, 2004), and providing a common cognitive and behavioral baseline that makes the tie more salient and rewarding to maintain (DiMaggio, 1987).~~

When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the cultural matching operates via two complementary interactional mechanisms, one affective and the other cognitive. First, drawing on Collins’s (2004) theory of interaction ritual chains, cultural matching provides the mutual conversational focal points necessary to produce successful interaction rituals (Meuleman, 2021). When two individuals share an interest in a cultural domain—such as music, films, sports, or recreational hobbies—they can readily initiate conversation, establish shared attention, and experience mutual entrainment. These successful interaction rituals generate positive emotional energy and a sense of solidarity, making the relationship intrinsically rewarding and motivating future interactions. Conversely, when cultural matching is absent, interactions require conscious cognitive effort, feel awkward or draining, and fail to generate emotional energy, leaving the tie vulnerable to benign neglect and decay.

Second, cultural matching increases the *cognitive salience* of an alter within an ego’s active network (Burt, 2000). Therefore, we expect that ~~commonalities in cultural tastes and leisure practices will significantly reduce~~ personal community (DiMaggio, 1987; Burt, 2002). In high-density environments like a college campus, individuals encounter numerous casual acquaintances. Alters with whom an ego shares specific cultural pursuits remain top of mind because they serve as immediate partners for concrete leisure activities (e.g., attending concerts, playing video games, or discussing books). This cognitive accessibility makes it more likely that the alter is regularly included in social plans, protecting the tie from the out-of-sight, out-of-mind dynamics that drive relational decay. Therefore, I anticipate that dyadic cultural matching significantly reduces the hazard of tie decay, net of ~~subjective closeness and structural controls~~ (structural embeddedness and dyadic controls (*The Baseline Matching Hypothesis 1*)).

2.3 Cultural Network Opacity as an Interactional Liability

If ~~shared cultural knowledge stabilizes ties, then a~~ cultural knowledge of an alter protects ties from decay, ~~then the~~ lack of ~~mutual cultural knowledge—it~~ cultural network opacity—should act as ~~a major, independent mechanism of an~~ independent liability that accelerates tie decay. We define cultural network opacity as the degree to which an ego is ignorant of an alter’s cultural preferences. ~~It is important, indicated empirically by explicit “Don’t Know” reports across cultural domains when asked to report on alter’s preferences.~~

It is conceptually crucial to differentiate opacity from ~~mere “taste dissimilarity.”~~ A ~~taste~~ cultural mismatch. A cultural mismatch implies that ego and alter know each other, the ego can identify both their own and their alter’s preferences but do not share them. Opacity, in contrast, implies, yet the two diverge (e.g., the ego reports enjoying sports

while reporting that the alter does not). Such mismatches still represent an informed relational connection where boundaries are known, at least from the ego's vantage point. Cultural network opacity, by contrast, signifies a relational void. When an ego reports "Don't Know" regarding the fact that ego cannot identify an alter's preferences across multiple domains, it signals a profound basic leisure domains reveals a meaningful lack of interactional investment. Furthermore, it creates a severe cognitive barrier to finding focal points for shared activities. Without a map of the alter's cultural landscape, the ego cannot easily initiate shared rituals or conversations, independently predicting a higher hazard of tie dissolution (Burt, 2002; Wellman et al., 1997). We test both the main effect of depth and personal disclosure (Wellman et al., 1997). This lack of cultural knowledge on the ego's part can render the relationship interactionally inert and less affectively salient, increasing the probability that the tie will be dropped from the ego's active network. Consequently, I hypothesize that cultural network opacity on tie decay and the moderation of the protective effect of known cultural matches by opacity independently increases the hazard of tie decay (*Hypothesis 2*).

2.4 Tie Strength and the Limits of Cultural Matching Contingencies: Strong-Tie Maintenance vs. Weak-Tie Scaffolding

Furthermore, recent research highlights a critical theoretical tension regarding how the effects of cultural matching intersect with tie strength. A central question concerns how the protective capacity of cultural matching intersects with tie strength. Existing literature suggests two competing predictions regarding this contingency.

On the one hand, drawing on the work of Edelman and Vaisey (2014) and Vaisey and Lizardo (2010), one might expect that cultural matching is primarily associated with the maintenance of culture and networks research (Vaisey and Lizardo, 2010; Edelman and Vaisey, 2014), cultural matching may be essential primarily for maintaining strong ties. In this view, already grounded friendships characterized by subjective closeness. From this perspective, intimate, emotionally significant friendships require ongoing cultural resonance and shared practices to sustain intimacy over time, making close ties alignment to sustain deep conversation, mutual understanding, and authentic rapport. Because strong ties carry high cognitive and emotional stakes, any perceived cultural misalignment or opacity becomes salient and disruptive. Consequently, strong ties should be highly sensitive to cultural matching (as a protective factor against decay (*The Hypothesis 3a: Strong-Tie Matching Hypothesis*)).

On the other hand, a competing perspective, drawn from structural network theory, suggests that strong ties are highly resilient due to strong ties may be inherently durable due to accumulated emotional inertia, multiplexity, and structural institutional commitments, and dense third-party embeddedness, making them less dependent on specific cultural similarities for survival resilient to differences in cultural tastes. Under this view, cultural matching acts as essential scaffolding specifically serves as essential compensatory scaffolding, particularly for weak or nascent newly formed ties. Weak ties suffer from a high baseline liability of newness (Burt, 2000); without shared interests the buffer of deep intimacy, they require immediate cultural commonalities to grease the wheels of interaction and provide focal points for engagement, they rapidly decay justify continued contact. Once a tie matures

into a close bond, cultural commonalities may become secondary to generalized interpersonal trust and shared history. Thus, the protective power of ~~matching~~ cultural matching against tie decay should be most pronounced ~~for among~~ weak ties (*The Hypothesis 3b: Weak-Tie Scaffolding Hypothesis*). ~~We estimate interaction models to adjudicate between these competing predictions~~

2.5 Positional Context, Institutional Structure, and Capital Conversion

Cultural choices and relational maintenance do not occur in an institutional or structural vacuum; instead, they happen within specific tie-formation foci and local network ecologies (Feld, 1982; McFarland et al., 2014). Actors occupy specific positions within campus social structures, defined by residential arrangements, interactional foci, and exogenous sociodemographic identities. Incorporating positional differences is therefore essential for avoiding an essentialized view of cultural matching and for understanding the conditions under which cultural capital translates into relational durability.

First, physical and institutional foci strongly condition the opportunity structure for interaction. In an undergraduate setting, residential arrangements (such as living in the same dormitory) provide recurring, low-cost opportunities for face-to-face contact. Shared spatial proximity can either amplify or substitute for cultural matching: roommates and dormmates may maintain ties through spatial convenience even in the absence of cultural matching, or shared living spaces may provide the necessary stage where cultural matching can be discovered and enacted.

Similarly, structural embeddedness—the extent to which an alter is connected to other members of an ego’s personal network (triadic closure)—provides reputational monitoring, collective validation, and group-level scaffolding that protects ties from decay (Granovetter, 1973; Coleman, 1988; Burt, 2000). Controlling for dyadic common alter counts and residential proximity is therefore critical to determining whether cultural matching has independent predictive power net of structural integration.

Second, drawing on Bourdieu’s (1984; 1986) theory of capital conversion, the capacity of cultural matching to convert into social capital (durable social ties) is intrinsically bounded by actors’ structural and institutional locations. Cultural capital does not convert into relational durability at an invariant exchange rate; its efficacy depends on the volume and composition of other capitals and the specific field conditions of the campus environment. Rather than operating as an unconstrained universal solvent that effortlessly dissolves social boundaries, cultural matching operates as a micro-interactional resource that functions *within* the constraints imposed by exogenous demographic categories (such as race and gender) and institutional opportunity structures (such as residential arrangements).

2.6 Summary of Hypotheses

In sum, ~~we test three core hypotheses regarding the role of culture in network evolution: I~~ evaluate four empirical predictions concerning the link between cultural matching, structure, and tie decay:

- ~~Baseline Matching Hypothesis: Shared cultural tastes and activities significantly reduce the hazard of tie decay~~Hypothesis 1 (Cultural Matching): Dyadic cultural matching significantly reduces the hazard of tie decay, net of structural embeddedness, residential proximity, and sociodemographic controls.
- ~~Cultural Network Opacity Hypothesis: A lack of mutual cultural knowledge independently increases the hazard of tie decay~~Hypothesis 2 (Cultural Network Opacity): Cultural network opacity independently increases the hazard of tie decay.
- ~~Moderation by Tie Strength: We test two competing predictions regarding moderation:—~~
- ~~H3a (Strong-Tie Matching): The protective effect of cultural matching is amplified among subjectively close ties.~~Hypothesis 3a (Strong-Tie Matching): The protective effect of cultural matching against tie decay is amplified among subjectively close (strong) ties.
- ~~H3b (Weak-Tie Scaffolding): The protective effect of cultural matching is amplified among weak ties to overcome the liability of newness~~Hypothesis 3b (Weak-Tie Scaffolding): The protective effect of cultural matching against tie decay is amplified for weak (non-close) ties, thereby overcoming the liability of newness.

3 Data and Methods

Data for this analysis comes from the *NetSense* study (Striegel et al., 2013), a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students’ social networks, demographic background, and cultural preferences.¹ ~~We collected~~The survey gathered information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended ~~activity nominations~~nominations of activities. For the closed-form items, egos were asked to rate both their own and their alters’ level of interest across six broad cultural and leisure domains: *music*, *movies*, *books*, *sports*, *games*, and *outdoor activities*. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity. Surveys were administered at the start and end of each semester via ~~Qualtrics during the study period~~Qualtrics.

3.1 ~~Study Waves~~Discrete-Time Event History Setup, Risk Set, and Tie Sequences

~~For this analysis of tie persistence, we pool data across adjacent waves to construct a~~To investigate tie decay over time, I construct a discrete-time person-period dataset spanning adjacent survey waves (Allison, 1982; Singer and Willett, 1993)—for a recent

¹~~A comprehensive codebook, data dictionary, and repository for the NetSense project are available at—~~

application of this approach to the study of tie evolution, see Kim et al. (2024). To isolate peer dynamics, ~~we restrict our~~ I restrict the sample to reported non-kin, on-campus ties (~~i.e. that is,~~ alters who are also Notre Dame students). ~~To capture relationship dynamics throughout the college experience, we construct a discrete-time person-period dataset spanning all available adjacent survey waves (Wave 1 through Wave~~ The *NetSense* study administered 8 survey waves across the students' college careers. Information on ego cultural preferences across the six closed-form domains was gathered across Waves 1 through 6, enabling the empirical modeling of tie decay across six adjacent wave transition intervals ($t = 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 5, 5 \rightarrow 6$, and $6 \rightarrow 7$). This ~~design allows us to observe tie persistence continuously across varying longitudinal~~ design allows for observing tie decay continuously across distinct developmental intervals of the ~~students' academic careers~~ undergraduate trajectory, capturing both the initial network churn of freshman year and the consolidation of ~~networks approaching graduation~~ personal networks as students approach upper-level student status.

The unit of analysis is the dyad-period (i, j, t) , defined for each alter j nominated in ego i 's active interaction roster at wave t . At each wave transition interval $t \rightarrow t + 1$, an active tie is at risk of experiencing tie decay. The dependent variable, Y_{ijt} , is a binary indicator scored as 1 if the tie is nominated again by ego i at wave $t + 1$ (indicating that the tie is protected from tie decay) and 0 if the alter is omitted from the ego's nomination roster at wave $t + 1$ (indicating tie decay). In discrete-time event history terms, the conditional hazard of tie decay at interval t is $h_{ijt} = P(Y_{ijt} = 0 \mid \text{active at } t)$, and the complementary probability of protection from tie decay is $1 - h_{ijt} = P(Y_{ijt} = 1 \mid \text{active at } t)$. In the analytic sample, 29.5 % of dyad-periods persist into the subsequent wave, reflecting the substantial baseline rate of tie decay characteristic of college peer networks.

A methodological question arises regarding the combinatorial trajectories of ties across multiple survey waves and the treatment of intermittent ties (ties that disappear and subsequently reappear). Across 8 survey waves, a dyad's nomination trajectory could theoretically exhibit any of $2^8 = 256$ binary presence/absence combinations ($2^7 = 128$ across the 7-wave analytical window).

In the *NetSense* data, however, only 151 distinct sequence configurations are observed. More importantly, the vast majority of dyads—79.8 % (4,716 of 5,908 unique dyads)—exhibit strictly contiguous relational histories: they are nominated in a single wave or persist continuously across consecutive waves until permanent decay or right-censoring. Only 20.2 % of dyads (1,192 dyads) ever exhibit an intermittent pattern in which a tie is omitted in one wave and re-nominated in a later wave. Within the complete-case analytic sample ($N = 5,349$ dyad-periods across 3,770 dyads), 84.5 % ($N = 4,522$) belong to the dyad's *first continuous spell* (from initial nomination until first decay or right-censoring), while only 15.5 % ($N = 827$) belong to recurrent spells.

Note that under this discrete-time framework, *rekindling does not count as persistence*. If an alter is nominated at wave t but omitted at wave $t + 1$, that dyad-period is coded as experiencing tie decay ($Y_{ijt} = 0$). If that alter is nominated again at a subsequent wave (for example, at wave $t + 2$), the discrete-time repeated-events model treats this re-appearance as the inception of a new, distinct risk episode (spell 2) covering the transition from $t + 2 \rightarrow t + 3$ (Allison, 1982; Box-Steffensmeier and Jones, 2004). Rekindling at wave $t + 2$ never retroactively alters the decay event observed at interval $t \rightarrow t + 1$. All time-varying

predictors (subjective closeness, contact frequency, cultural matching, and structural embeddedness) are dynamically re-considered at wave $t + 2$ as predictors of the subsequent outcome at wave $t + 3$. To confirm that the findings do not depend on the inclusion of recurrent spells, I report a sensitivity analysis in Section 4.3 that strictly enforces an *absorbing first-decay* specification, tracking ties exclusively until their first decay event and dropping all subsequent rekindled spells ($N = 4,522$).

A related consideration concerns *node persistence* (ego survey retention) and the threat of survivorship or panel attrition bias. In the *NetSense* panel, respondents completed an average of 5.12 survey waves (median 5 waves), with 80.3 % (151 of 188 egos) participating in 4 or more waves and 48.9 % completing 6 or more waves. In discrete-time event-history setups, when an ego does not complete survey wave $t + 1$, their active ties from wave t are formally right-censored at the interval $t \rightarrow t + 1$. Auxiliary attrition regressions confirm that ego panel retention is completely uncorrelated with baseline cultural matching: neither closed-form matching ($p = 0.98$) nor open-ended matching ($p = 0.23$) predicts total waves completed or study dropout. To guard against potential selection biases, Section 4.3 evaluates sensitivity models restricted to complete wave-to-wave transitions and high-retention egos.

3.2 Measures and Descriptive Statistics

Tables 1 and 2 present summary statistics for all continuous and categorical variables used in our analyses across the $N = 5,584$ dyad-period observations.

Table 1: Descriptive Statistics (Continuous Variables)

	Mean	SD	Min	Max
Closed-Form Cultural Matching	1.67	1.49	0.00	6.00
Open-Ended Activity Matching	2.52	1.34	0.00	5.00
Cultural Network Opacity	0.45	1.13	0.00	6.00
Structural Embeddedness (Common Alters)	5.74	4.18	0.00	19.00
Tie Duration (Years)	4.23	6.21	0.02	21.16

3.2.1 Outcome Variable: *Protection from Tie Decay*

Protection from Tie Persistence/Decay: A Network evolution is evaluated via tie decay between adjacent waves. As detailed above, the outcome is measured as a binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2; *protected from tie decay*), and 0 if the tie has dissolved (i.e. decayed (that is, the alter is no longer listed in the ego’s network)). In our analytic sample, 29.5 % of dyad-periods persist into the subsequent wave, reflecting the substantial baseline turnover characteristic of collegiate peer networks.—To determine tie presence, egos were asked to list the people with whom they interact most frequently, using the following name generator: “Please list the people with whom you have spent the most time in the last month.”—Egos could nominate up to 20 alters in each wave. This

Table 2: Descriptive Statistics (Categorical Variables)

Variable	Category	N	Percent
Tie Persisted (Protection from Tie Decay)	0 (Decayed)	3935	70.5%
	1 (Persisted)	1649	29.5%
Ego Gender Identity	Men	2901	54.0%
	Women	2471	46.0%
Alter Gender Identity	Men	2839	51.0%
	Women	2728	49.0%
Subjective Closeness	Not Close	565	10.1%
	Somewhat Close	2611	46.8%
	Close	2400	43.0%
Interaction Frequency	Weekly	760	13.6%
	Daily	4824	86.4%
Race Homophily	Heterophilous	510	9.1%
	Homophilous	1508	27.0%
	Unknown	3566	63.9%

name generator uses what (Marin and Hampton, 2007) ~~refers to as~~ calls the “interaction” approach, which is relatively neutral with respect to tie strength (Laumann et al., 1989) (Laumann et al., 1989; Marsden and Campbell, 1984). Other tie characteristics related to strength, such as duration, role relation (e.g., kin versus friend), and subjective closeness, were collected via follow-up name interpreters.

3.2.2 ~~Main Cultural~~ Predictor~~of Interest~~

Closed-Form Cultural Matching: A count variable (ranging from 0 to 6; Mean = 1.67, SD = 1.49) indicating the number of closed-form cultural domains in which the ego and alter ~~share similar tastes~~ exhibit cultural matching. Egos were asked to rate their own interest, and their alters’ interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey ~~prompt for egos was:~~ prompts were: for egos, “How interested are you in the following activities?” and for alters: ~~–,~~ “How interested is this person in the following activities?” Response categories included “Very interested,” “Somewhat interested,” and “Not at all interested.” A domain was scored as a “match” if the ego rated both themselves and their alter identically across these categories. Note that the ego’s information was collected in a *separate* survey from the ego-network survey (a demographic survey), which dealt with immediate ordering bias and attenuated any cognitive consistency effects. As evaluated in Section 4.3, positive interest alignment (shared interest ratings of “Very interested” or “Somewhat interested”) accounts for 92.0 % of all observed closed-form matches.

Open-Ended Activity Matching: A count variable (ranging from 0 to 5; Mean = 2.52, SD

= 1.34) capturing the ~~number of shared specific~~ degree of cultural matching across specific leisure activity nominations. Egos were asked to freely list up to ~~five~~ 5 leisure activities they enjoyed ~~with~~, using the prompt: “Please list up to 5 specific activities that you enjoy doing in your free time.” Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed, prompted with: “Which of your favorite activities does this person also enjoy?”

Cultural network opacity: A count variable (ranging from 0 to 6; Mean = 0.45, SD = 1.13) measuring the number of cultural domains for which the ego responded “Don’t know” regarding their alter’s preferences. For the closed-form domain questions regarding alters, respondents were provided a fourth response category: “Don’t know.” While survey researchers often instinctively treat “Don’t know” responses as missing data to be imputed or dropped, ~~we~~ I explicitly operationalize these responses as substantive data. In the context of ego-network surveys, a “Don’t know” response is not random missingness; rather, it is a direct behavioral measure of relational ignorance (opacity). By retaining and counting these responses, ~~we~~ I treat opacity as a distinct structural feature of the dyadic tie.

3.2.3 ~~Additional Predictors~~ Longitudinal Stability and Consistency of Cultural Tastes

A fundamental premise in theories of cultural capital and relational selection is that cultural tastes represent relatively durable personal orientations rather than ephemeral, fast-shifting pursuits (Lizardo, 2006). If students’ cultural tastes fluctuated erratically across semesters, or if observed dyadic similarity arose primarily because connected friends rapidly adopt and abandon new lifestyle practices in lockstep, cultural matching could not serve as an enduring micro-interactional anchor protecting ties from decay (Vaisey and Lizardo, 2010).

To evaluate the empirical durability of cultural tastes in this undergraduate population, Table 3 reports longitudinal consistency metrics across the panel. First, I estimate two-level random-intercept models ($N = 848$ observations across 170 egos) to decompose between-person and within-person variance across Waves 1–3 and calculate the Intraclass Correlation Coefficient (ICC). The estimated ICCs indicate substantial taste stability: Engaging in sports exhibits an ICC of 0.80, reading and book interests 0.70, outdoor activities 0.61, movies 0.57, video games 0.54, and music 0.43 (the lower variance in music likely reflects ceiling effects in this population, where over 75 % of respondents consistently select “Very much”).

Second, adjacent wave-to-wave test-retest correlations (r_{adj}) average between 0.50 and 0.80, with exact response agreement ranging from 61.4 % to 78.5 % and agreement within plus or minus one category exceeding 96.6 % to 99.7 % across all domains. Multi-year test-retest correlations spanning 1.5 years (Wave 1 to Wave 3; r_{W1-W3}) range from 0.51 to 0.81 for games, movies, outdoor activities, books, and sports. Third, for persisting ties, egos’ longitudinal perceptions of their alters’ tastes exhibit strong test-retest reliability ($r_{alter} = 0.54$ to 0.77), with 94.6 % to 98.1 % of perceptions remaining within plus or minus one scale step across consecutive waves.

Table 3: Longitudinal Stability and Test-Retest Consistency of Cultural Tastes across Waves

Domain	Ego Self-Reported Tastes					Alter Perceived Tastes	
	ICC	r_{adj}	Exact (%)	± 1 (%)	$r_{\text{W1} \rightarrow \text{W3}}$	r_{alter}	± 1 (%)
Sports	0.80	0.80	74.4	98.8	0.81	0.77	96.2
Books	0.70	0.72	76.9	99.1	0.65	0.70	95.8
Outdoor	0.61	0.62	68.8	98.8	0.62	0.63	96.0
Movies	0.57	0.57	76.3	99.4	0.62	0.54	97.8
Games	0.54	0.55	61.4	96.6	0.51	0.54	94.6
Music	0.43	0.50	78.5	99.7	0.36	0.64	98.1

^a ICC (Intraclass Correlation Coefficient) estimated from two-level random intercept models decomposing between-ego versus within-ego variance across Waves 1–3. r_{adj} , Exact, and ± 1 report mean adjacent wave-to-wave test-retest correlations and agreement percentages. $r_{\text{W1} \rightarrow \text{W3}}$ reports multi-year stability across a 1.5-year span. Alter stability metrics are computed across consecutive waves for persisting dyads ($N = 2,546$).

3.2.4 Relational and Structural Controls

Subjective Closeness: Measured with the name interpreter, “How close are you to this person?” Response categories were “Not close” (10.1 %), “Somewhat close” (46.8 %), and “Very close” (43.0 %). This is treated as a factor variable in the models. As Marsden and Campbell (1984) noted in their classic paper, this item is the most reliable indicator of tie strength (Granovetter, 1973).

Relationship Type: Derived from the role relation name interpreter “How are you connected to this person?” We include binary indicators for whether the tie is a Friend (vs. other types) or a Roommate/Dormmate (capturing on-campus structural propinquity).

Frequency of Interaction: Measured with the name interpreter, “How often do you communicate with this person?” Responses were dichotomized into a Daily contact indicator (86.4 %) versus a reference category of Weekly contact (13.6 %).

Tie Duration: A continuous measure of how long the ego has known the alter (in years; Mean = 4.23 years, SD = 6.21), asked via the name interpreter: “How long have you known this person?” We include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.

Structural Embeddedness (Common Neighbors / Triadic Closure): To account for the collective, group-embedded nature of personal networks and control for triadic closure (Granovetter, 1973; Coleman, 1988), we construct a dyadic embeddedness metric from the perceived alter-by-alter acquaintance matrix. In each survey wave, respondents evaluated whether their nominated contacts knew one another (“As far as you know, does [Alter Name] know any of your other contacts that are listed below?”). From this dyadic matrix, we compute the number of common neighbors (shared contacts) that Alter j possesses with other

alters in ego’s network (Mean = ~~5.43~~5.74, SD = ~~4.29~~4.18; ranging from 0 to 19). Because the ego is tied to every alter ~~in the roster~~, each reported alter-alter connection directly represents a closed triangle (Ego – Alter_j – Alter_k). In ~~our~~the models, this measure is standardized. For wave intervals where the alter-alter matrix was not included (Wave 6), the variable is standardized using a ~~non-parametric wave transition~~nonparametric wave transition fixed effect to absorb the baseline shift, and sensitivity checks are conducted on the fully measured waves.

Demographics: ~~We-I~~I control for ego’s gender identification (~~44.3 % Women, 52.0 % 46.0 % Women, 54.0 %~~ Men) and perceived (from ego’s point of view) alter gender (~~48.9 % Women, 50.8 % 49.0 % Women, 51.0 %~~ Men), along with a multiplicative interaction term to capture same-gender vs. cross-gender homophily and heterophily (respectively). ~~We-I~~I also include a measure of *Race Homophily*, constructed by matching the ego’s self-reported race to the alter’s perceived race (27.0 % Homophilous, 9.1 % Heterophilous, 63.9 % Unknown alter race).

Time Period: Dummy variables for each wave transition interval (Wave 1 to 2, Wave 2 to 3, Wave 3 to 4, Wave 4 to 5, Wave 5 to 6, Wave 6 to 7, Wave 7 to 8) to allow the baseline hazard of tie ~~persistence~~decay to vary non-parametrically across the entire panel span.

3.3 Analytical Strategy ~~and Model Specification~~

Because ~~our data tracks the data track~~ relationships over time and individuals can nominate multiple alters, ~~our the~~ data structure consists of dyad-periods nested within egos. To properly analyze tie ~~persistence~~decay, capture the high baseline volatility of personal networks (Cornwell et al., 2021; Weiss et al., 2022), and account for right-censoring, ~~we-I~~I employ a discrete-time event history framework (Singer and Willett, 1993). Since the outcome ~~variable—whether a tie persists—variable—whether a tie is protected from decay~~ into the next wave—is binary, we estimate the hazard of tie survival (the probability of persistence given the tie survived to wave—is binary, I estimate protection from tie decay (the conditional probability of tie persistence given that the tie was active in the previous wave) using mixed-effects logistic regression.

This modeling strategy allows ~~us-me~~me to explicitly model the baseline hazard rate of tie decay across each wave transition, isolating the protective effects of ~~our cultural variables~~cultural matching against this background volatility. ~~Furthermore, because the discrete-time event history framework dynamically measures cultural matching at wave t to predict the hazard of tie decay into wave $t + 1$, the analytical model does not presuppose taste invariance within the observation period; rather, it evaluates whether contemporaneous cultural matching at the start of each transition interval protects the tie over the subsequent period.~~

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), ~~we-I~~I include random intercepts for egos (Agresti, 2007). This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1 | \text{active at } t)) = \beta_0 \alpha_t + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where Y_{ijt} is the ~~indicator of tie persistence~~ binary indicator of protection from tie decay (1 = persisted into wave $t + 1$; 0 = tie decay) for ego i and alter j at time t , α_t represents the period-specific baseline hazard intercept (captured via wave transition fixed effects), X_{ijt} represents the vector of ~~our core theoretical predictors~~ (cultural matching core cultural matching predictors (closed-form cultural matching, open-ended activity matching, and cultural network opacity), C_{ijt} is a vector of time-varying dyadic and structural control variables (including ~~time period indicators to allow the baseline hazard to vary non-parametrically over waves~~ subjective closeness, contact frequency, relationship type, race homophily, gender composition, tie duration, and structural embeddedness), and u_i is the ego-specific random intercept, assumed to be normally distributed with mean zero and variance estimated from the data ($u_i \sim \mathcal{N}(0, \sigma_u^2)$).

Because the sample was recruited from an undergraduate student cohort, a small proportion of nominated alters are also study participants (egos). In the full analytic panel of $N = 5,584$ dyad-periods (3,134 unique alters across 182 unique egos), exactly 102 unique alters (3.3 % of all nominated alters, representing 7.0 % of dyad-period observations) are also egos in the sample. Within the complete-case sample used for model estimation ($N = 5,349$ dyad-periods across 180 egos), this overlap corresponds to 374 dyad-periods (also 7.0 %). To ensure that this partial overlap does not violate the standard hierarchical nesting assumption, I also estimate *cross-classified multilevel models* (CCMM) with two-way crossed random effects:

$$\text{logit}(P(Y_{ijt} = 1 \mid \text{active at } t)) = \alpha_t + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i + v_j \quad (2)$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ captures ego-specific heterogeneity and $v_j \sim \mathcal{N}(0, \sigma_v^2)$ captures alter-specific popularity or retention propensity, alongside models clustering on unique undirected dyads ($i \leftrightarrow j$).

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package (Bates et al., 2015) in R (R Core Team, 2026). In models that interact subjective closeness with ~~our cultural variables, we~~ cultural matching variables, I calculate and plot predicted probabilities to facilitate the interpretation of the non-linear interaction terms (Mize, 2019).¹ In addition, to ensure that findings are not sensitive to the multi-spell setup, two-way dyadic clustering, or between-ego selection, I evaluate three complementary sensitivity checks: an *absorbing first-decay* mixed-effects model restricting follow-up strictly to the initial continuous tie spell, *cross-classified and dyadic random effects models*, and an *ego fixed-effects* conditional logit model isolating within-ego variation.

4 Results

~~We present our~~ I present the findings in three stages. First, ~~we~~ I examine the baseline main effects of cultural matching and opacity on ~~tie persistence~~ protection from tie decay. Second,

¹To verify the robustness of our core findings to modeling specification choices, we conduct a supplementary analysis by substituting our mixed-effects modeling framework with an ego fixed-effects framework using conditional logistic regression (Gaubard and Korn, 2011). This adjusts for all unobserved, time-invariant ego-level characteristics.

~~we test the Strength-Mediated Matching Hypothesis by interacting our~~ I test the competing moderation predictions (Hypotheses 3a and 3b) by interacting matching variables with subjective closeness. Finally, ~~we assess the robustness of our findings using ego fixed-effects~~ I evaluate the sensitivity of the findings by considering different specifications of the tie decay process that eliminate the possibility of tie rebirth, statistically evaluating the issue of egos who are also alters of other egos, accounting for survivorship bias due to differential sample retention of egos, and adjusting for all time-invariant ego-level characteristics via fixed effects (conditional logit) models.

4.1 Main Effects: ~~Cultural Matching~~, ~~Cultural Network Opacity~~, and ~~Structural Embeddedness~~

Table 4 presents the odds ratios from a series of mixed-effects logistic regression models predicting ~~tie survival~~ protection from tie decay. As standard network theory would predict, ties that are subjectively closer ($OR = 2.39, p < 0.001$ for “Close” ties in Model 1) are significantly more ~~likely to survive~~ protected from tie decay. Interestingly, having daily contact ($OR = 1.12, p > 0.05$) does not reach conventional significance levels when adjusting for closeness and other controls in this sample.

Building on this structural baseline, Model 1 shows that *closed-form cultural matching* significantly ~~increases the odds of tie persistence~~ protects ties from decay. For each additional broad cultural domain in which an ego and alter ~~share tastes, the tie is 9.6 % more likely to survive into the subsequent wave~~ cultural match, the odds of protection from tie decay increase by about 9.6 % ($OR = 1.10, p < 0.01$), net of controls.

In Model 2, ~~we introduce~~ I introduce *open-ended activity matches* matching. ~~Specific, shared behavioral practices provide additive protective value~~ Cultural matching in specific leisure activities provides additional protective value against tie decay: each shared activity increases ~~the odds of tie survival~~ protection against tie decay by 8.0 % ($OR = 1.08, p < 0.05$). When both are included, closed-form cultural matching remains significant ($OR = 1.08, p < 0.05$), indicating that specific ~~activities provide~~ activity matching provides protective value beyond broad ~~categorical taste similarities~~ domain matching.

Model 3 tests the independent main effect of *cultural network opacity*—measured as the number of domains in which the ego reported “Don’t Know” regarding their alter’s preferences. ~~Opacity I find that opacity~~ predicts a 5.3 % ~~decrease in the odds of tie persistence~~ increase in the hazard of tie decay per unknown domain ($OR = 0.95$ for protection from decay). However, this main effect is not statistically significant at ~~the conventional level~~ conventional levels ($p > 0.05$) when adjusting for both ~~closed~~ closed-form and open-ended ~~matches~~ cultural matching simultaneously. The attenuation of this coefficient in the pooled mixed-effects model is likely due to the built-in, between-person correlation between reporting “Don’t Know” and having fewer known ~~closed-form~~ cultural matches. However, as ~~we will~~ I show in the interaction models and the fixed-effects ~~robustness~~ sensitivity check below, opacity remains a ~~significant liability in specific relational contexts~~ substantively significant liability for tie durability when considering only within-ego variance.

Finally, Model 4 introduces ~~structural embeddedness~~ structural embeddedness—the standardized count of common neighbors (shared alter contacts) that an alter possesses

within ego’s nominated network. Consistent with triadic closure and ~~collective embeddedness arguments~~ dynamic Simmelian tie arguments (Burt, 2000; Krackhardt and Handcock, 2007), having shared network contacts strongly and significantly ~~bolsters tie persistence~~ protects ties from decay ($OR = 1.12, p < 0.05$): each standard deviation increase in the number of common alter contacts increases ~~the odds of tie survival~~ protection from tie decay by 12.2 %. ~~Crucially, controlling~~

Importantly, adjusting for structural embeddedness does not eliminate the independent contribution of cultural matching. Both open-ended ~~matches activity matching~~ ($OR = 1.07, p < 0.05$) and closed-form cultural ~~matches matching~~ ($OR = 1.07, p = 0.057$) maintain their positive protective associations ~~.—This demonstrates against tie decay. This shows~~ that dyadic cultural ~~affinity matching~~ is not an artifact of being embedded in dense peer cliques; rather, ~~culture cultural matching~~ and triadic structure operate as complementary, parallel anchors ~~of tie durability~~. ~~Supplementary tests interacting cultural matching with protecting ties from decay. Supplementary models examining the interaction between cultural matching and~~ structural embeddedness revealed no significant moderation ($p = 0.65$), indicating that ~~shared tastes protect ties across both structurally peripheral and densely embedded relationships~~ cultural matching protects ties from decay for both Simmelian and non-Simmelian ties.

To contextualize these coefficients, Figure 1 plots the average marginal effect of each core theoretical variable on the probability of ~~tie persistence~~ protection from tie decay. Across the board, cultural matching serves as a protective mechanism against tie decay (positive estimates), whereas cultural network opacity ~~acts as a liability~~ accelerates tie decay (negative estimates).

4.2 ~~The Moderating Role of~~ Moderation by Subjective Closeness

While the main effects show that cultural ~~alignment generally stabilizes ties~~, we matching generally protects ties from decaying, I next assess whether ~~this “social glue” is uniformly necessary for all relationships. We~~ the protective effect is observed across all relationship types. To do so, I estimated auxiliary interaction models (not shown in the tables but visualized using predicted probabilities) to test whether the protective effects of cultural matching and the detrimental effects of cultural network opacity are moderated by baseline subjective closeness. Because interaction terms in logistic models can be opaque due to the non-linear link function, ~~we I~~ visualize this critical interaction to make the core finding ~~instantly digestible~~ directly interpretable. Figure 2 plots the average marginal effects of ~~our the~~ three cultural variables across ties rated as “Not Close,” “Somewhat Close,” and “Close.”

The results reveal a structured asymmetry in ~~the protective effects of how~~ cultural matching and ~~the deleterious effect of opacity~~ cultural network opacity protect ties from decay conditioned on tie strength. ~~We I~~ find a strong interaction for both closed-form cultural ~~matches matching~~ and cultural network opacity, but little interaction for open-ended ~~matches activity matching~~. For broad, closed-form ~~matches cultural matching~~ and network opacity, the predictive slopes (marginal effects) are steepest and most significant for *strong* ties (those rated “Close”). For these ~~essential stronger~~ relationships, an increase in ~~shared cultural tastes dramatically~~ cultural matching substantially increases the predicted probability of ~~tie persistence~~ protection from tie decay. Correspondingly, cultural network opacity ~~has imposes~~

Table 4: Odds Ratios for Protection from Tie Decay (Main Effects Models)

	Model 1 Closed	Model 2 + Open	Model 3 + Opacity	Model 4 + Embed.
Closed-Form Matches	1.096** (0.035)	1.083* (0.035)	1.070* (0.036)	1.067+ (0.036)
Open-Ended Matches		1.080* (0.034)	1.076* (0.034)	1.071* (0.033)
Network Opacity			0.947 (0.038)	0.952 (0.038)
Structural Embeddedness (Common Alters)				1.122* (0.050)
Subjective Closeness: Somewhat	1.440** (0.203)	1.388* (0.197)	1.368* (0.195)	1.342* (0.191)
Subjective Closeness: Close	2.388*** (0.359)	2.244*** (0.342)	2.188*** (0.336)	2.097*** (0.324)
Roommate/Dormmate	0.788 (0.195)	0.787 (0.194)	0.783 (0.194)	0.791 (0.197)
Friend	0.714 (0.175)	0.705 (0.173)	0.706 (0.174)	0.712 (0.177)
Frequency: Daily (vs Weekly)	1.119 (0.146)	1.107 (0.145)	1.085 (0.143)	1.081 (0.143)
Race Homophily: Same Race	0.950 (0.122)	0.944 (0.122)	0.935 (0.121)	0.932 (0.120)
Tie Duration (Scaled)	0.796 (0.183)	0.790 (0.182)	0.768 (0.178)	0.740 (0.172)
Tie Duration Sq (Scaled)	1.238 (0.303)	1.262 (0.309)	1.288 (0.317)	1.339 (0.331)
Num.Obs.	5349	5349	5349	5349
AIC	5147.9	5143.6	5143.7	5139.0
BIC	5273.0	5275.3	5282.0	5283.8

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models control for ego and alter gender, non-linear tie duration, and baseline hazard variations across all 7 wave transitions (coefficients omitted for space). Random intercepts for egos are included.

its most severe, statistically significant penalty precisely on these same strong ties: ~~a lack of mutual cultural knowledge~~ the ego’s lack of knowledge of the alter’s tastes significantly accelerates their decay for ties ~~that are~~ rated as subjectively close. Conversely, for weak ties (those rated “Not Close”), the marginal effects for both closed-form cultural matching and opacity are relatively flat, indicating an intrinsic ~~fragility to vulnerability to tie~~ decay regardless of cultural ~~alignment matching~~ or opacity in these broad domains.

This interaction suggests that ~~shared cultural tastes and mutual cultural knowledge~~ cultural matching and the ego’s cultural knowledge of the alter constitute the substantive “core” of ~~deep, institutionalized deeper~~ friendships, requiring ongoing cultural ~~alignment to be sustained~~ matching to protect them from decay. Weak ties, on the other hand, are fleeting and superficial; their baseline hazard of decay is so high that cultural matching does little to ~~save them~~ prevent their decay. In contrast, the protective effect of specific, open-ended activity ~~matches matching~~ operates more uniformly across relationships of all strengths, providing a consistent anchor ~~for tie persistence~~ protecting ties from decay, whether the tie is new or well-established.

4.3 ~~Robustness—Check~~ Sensitivity Analyses: Absorbing First Decay, Cross-Classified Models, and Ego Fixed-Effects ~~Fixed Effects~~

To ~~confirm that our~~ ensure that these empirical findings are not artifacts of our primary mixed-effects specification, we estimated an model specification, multi-spell assumptions, or clustering structures, I evaluate three complementary sensitivity checks.

First, in Model 1 of Table 7, I evaluate sensitivity to the handling of recurrent spells and tie rekindling by estimating an ~~Ego Fixed-Effects model~~ absorbing first-decay using conditional logistic regression mixed-effects logistic regression model ($N = 4,522$ complete dyad-periods across 3,770 dyads). Under this specification, each dyad is followed strictly until its first non-nomination event ($Y_{ijt} = 0$) or right-censoring; all subsequent recurrent spells following a temporary lapse are excluded from the risk set. As shown in the first column of Table 7, restricting the sample to the initial continuous spell yields results substantively equivalent to those of the primary models. Closed-form cultural matching remains a positive and statistically significant predictor of protection from tie decay ($OR = 1.086, p = 0.029$), structural embeddedness continues to strongly protect ties from decay ($OR = 1.120, p = 0.021$), and subjective closeness maintains its strong protective effect ($OR = 2.002, p < 0.001$ for “Close” ties). This indicates that the findings do not depend on the inclusion of recurrent or rekindled spells.

Second, to address potential non-independence arising from alters who are also study egos (or alters nominated across multiple egos), Table 5 reports estimates from cross-classified multilevel models with two-way (ego and alter) crossed random effects (Column 2) and undirected dyad clustering (Column 3, 3,708 unique dyad clusters within the complete-case sample). As shown in Columns 2 and 3 of Table 5, explicitly modeling crossed alter-level or dyad-level random effects leaves the point estimates and standard errors for the core predictors virtually identical to the standard hierarchical specification (Column 1): open-ended activity matching remains significant ($OR = 1.070\text{--}1.071, p < 0.05$), closed-form

cultural matching remains positive ($OR = 1.065-1.066, p = 0.06-0.07$), and structural embeddedness continues to strongly protect ties from decay ($OR = 1.120-1.121, p < 0.05$). Furthermore, completely excluding the 374 dyad-periods where alter is also a survey ego (Column 4, $N = 4,975$) yields identical conclusions ($OR = 1.069, p = 0.043$ for open matching; $OR = 1.140, p = 0.006$ for structural embeddedness).

Third, to guard against potential panel attrition or survivorship bias (node persistence), I re-estimated the models across restricted subsamples based on survey retention. When follow-up is strictly right-censored at the point of ego survey non-response (see Table 7), $N = 4,855$ complete wave-to-wave transitions), open-ended activity matching remains a strong and significant protector against tie decay ($OR = 1.094, p = 0.007$), structural embeddedness preserves tie durability ($OR = 1.087, p = 0.078$), and cultural network opacity accelerates tie decay ($OR = 0.928, p = 0.074$). Subsetting further to high-retention egos who completed 4 or more waves ($N = 4,605$ dyad-periods across 151 egos) and 6 or more waves ($N = 3,253$ dyad-periods across 92 egos) yields substantively identical results ($OR = 1.079, p = 0.028$ and $OR = 1.075, p = 0.076$ for open-ended matching, respectively).

Fourth, to evaluate whether the protective effect of cultural matching is driven by shared positive passions versus shared disinterest (negative homophily), Table 6 decomposes closed-form matching into positive interest alignment and shared disinterest. As shown in Model 2 of Table 6, exact positive interest matches significantly protect ties from decay ($OR = 1.072, p = 0.047$), and strong positive matches (Model 3, where the ego rates both themselves and their alter as “Very interested”) provide an even stronger protective association ($OR = 1.087, p = 0.036$). In contrast, shared disinterest (e.g., the ego rating both themselves and their alter as “Not at all interested” in a domain) exhibits no statistically significant association with tie protection ($OR = 1.075, p = 0.386$ in Model 2; $OR = 1.069, p = 0.426$ in Model 3). This indicates that active positive cultural choices serve as a relational anchor protecting ties from decay, complemented by open-ended activity matching, which, by design, captures shared participation in favorite leisure pursuits ($OR = 1.070, p = 0.030$).

Fifth, in Model 2 of Table 7, I estimate an *ego fixed-effects* model using conditional logistic regression (Gaubard and Korn, 2011) across $N = 1,640$ informative dyad-periods within egos exhibiting variation in tie decay. By stratifying the likelihood function by ego, this approach leverages only strictly within-ego variance—comparing variance—comparing the alters a single ego retains with those it drops. This more stringent test more effectively isolates dyadic tie characteristics from unobserved individual baseline differences (e.g., an ego’s stringent test effectively purges all unobserved, time-invariant ego characteristics (such as baseline sociability, introversion, or general propensity to use personal response tendencies toward the “Don’t Know” survey option).

As shown in the table, the fixed-effects models yield results that are remarkably consistent with our random-effects main models. Closed-form matches ($OR = 1.05, p < 0.10$) and open-ended matches ($OR = 1.06, p < 0.05$) remain positive predictors of tie persistence, with odds ratios that are very close in magnitude to the baseline models (second column of Table 7, the within-ego conditional logit estimates are substantively similar to the main mixed-effects models. Open-ended activity matching remains a statistically significant predictor of protection against tie decay ($OR = 1.056, p < 0.05$), while closed-form cultural matching remains positive ($OR = 1.046, p < 0.10$). Crucially, in this strict within-ego specification,

Table 5: Sensitivity Analysis: Cross-Classified Random Effects and Dyadic Clustering Models

	Model 1 Standard	Model 2 Crossed	Model 3 Dyad	Model 4 Excl. Ego-Alters
Closed-Form Cultural Matching	1.067+ (0.034)	1.065+ (0.034)	1.066+ (0.034)	1.062+ (0.036)
Open-Ended Activity Matching	1.071* (0.031)	1.071* (0.032)	1.070* (0.032)	1.069* (0.033)
Cultural Network Opacity	0.952 (0.040)	0.953 (0.041)	0.954 (0.040)	0.947 (0.042)
Structural Embeddedness (Common Alters)	1.122* (0.045)	1.121* (0.046)	1.120* (0.045)	1.140** (0.047)
Subjective Closeness: Somewhat	1.342* (0.143)	1.329* (0.145)	1.328* (0.144)	1.377* (0.150)
Subjective Closeness: Close	2.097*** (0.154)	2.071*** (0.157)	2.068*** (0.156)	2.161*** (0.163)
Roommate/Dormmate	0.791 (0.249)	0.798 (0.255)	0.791 (0.252)	0.732 (0.256)
Friend	0.712 (0.248)	0.708 (0.254)	0.713 (0.251)	0.740 (0.254)
Frequency: Daily (vs Weekly)	1.081 (0.132)	1.070 (0.134)	1.076 (0.133)	1.023 (0.138)
Tie Duration (Scaled)	0.740 (0.232)	0.745 (0.236)	0.742 (0.234)	0.666+ (0.242)
Tie Duration Sq (Scaled)	1.339 (0.247)	1.328 (0.252)	1.336 (0.250)	1.512 (0.256)
Ego Random Intercept SD (σ_u)	0.610	0.614	0.607	0.657
Alter/Dyad Random Intercept SD		0.427	0.328	
Dyad-Periods (N)	5349	5349	5349	4975
Unique Egos	180	180	180	180

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models control for ego and alter gender identity, gender interaction, race homophily, structural embeddedness, and wave transition fixed effects. Model 1 is the primary hierarchical random-intercept model. Model 2 estimates crossed random intercepts for both egos and alters. Model 3 specifies random intercepts for egos and unique undirected dyads. Model 4 excludes the 374 dyad-periods where alter is also a survey ego, within the complete-case modeled sample.

Table 6: Sensitivity Analysis: Decomposing Shared Positive Interests versus Shared Disinterest

	Model 1 Aggregated	Model 2 Exact Likes	Model 3 High Likes
Closed-Form Cultural Matching (Aggregated)	1.067+		
	(0.034)		
Shared Positive Interests (Exact)		1.072*	
		(0.035)	
Shared Strong Interests (High)			1.087*
			(0.040)
Shared Disinterest / Dislikes		1.075	1.069
		(0.084)	(0.083)
Open-Ended Activity Matching	1.071*	1.070*	1.068*
	(0.031)	(0.031)	(0.031)
Cultural Network Opacity	0.952	0.955	0.947
	(0.040)	(0.040)	(0.039)
Structural Embeddedness (Common Alters)	1.122*	1.122*	1.122*
	(0.045)	(0.045)	(0.045)
Subjective Closeness: Somewhat	1.342*	1.343*	1.335*
	(0.143)	(0.143)	(0.143)
Subjective Closeness: Close	2.097***	2.099***	2.064***
	(0.154)	(0.155)	(0.155)
Roommate/Dormmate	0.791	0.788	0.787
	(0.249)	(0.249)	(0.249)
Friend	0.712	0.710	0.714
	(0.248)	(0.248)	(0.248)
Frequency: Daily (vs Weekly)	1.081	1.078	1.081
	(0.132)	(0.132)	(0.132)
Tie Duration (Scaled)	0.740	0.736	0.732
	(0.232)	(0.232)	(0.232)
Tie Duration Sq (Scaled)	1.339	1.344	1.352
	(0.247)	(0.247)	(0.247)
Ego Random Intercept SD (σ_u)	0.610	0.608	0.609
Dyad-Periods (N)	5349	5349	5349
Unique Egos	180	180	180

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models include controls for ego and alter gender identity, gender interaction, race homophily, wave transition fixed effects, and ego random intercepts ($N = 5,349$ dyad-periods across 180 egos). Model 1 is the primary aggregated matching model. Model 2 decomposes closed-form matching into exact positive interest matches and shared disinterest. Model 3 isolates strong positive matches (ego rating both self and alter as “Very much”). All alter ratings are ego-reported perceptions, not direct alter self-reports.

cultural network opacity emerges as a highly significant ~~negative predictor of tie persistence~~ ($OR = 0.92, p < 0.01$) ~~accelerator of tie decay~~ ($OR = 0.920, p < 0.01$), reducing the odds of tie survival by 8.2 % ~~protection from tie decay by 8.0 %~~ per unknown domain. This ~~confirms~~ ~~indicates~~ that while opacity may correlate with overall ~~match rates across cultural matching rates between~~ individuals, *within* an individual’s personal network, ~~a lack of cultural knowledge about relational ignorance regarding~~ a specific alter is ~~a powerful, independent signal of tie dissolution.~~ ~~The structural~~ ~~an independent contributor to tie decay.~~ Dyadic controls—such as being ~~subjectively~~ “Close” ($OR = 1.99, p < 0.001$) ~~$OR = 1.949, p < 0.001$~~ and “Somewhat Close” ($OR = 1.409, p < 0.01$)—also maintain their ~~consistent associations with the baseline hazard of tie decay.~~ ~~strong protective associations against tie decay.~~

To better assess the magnitude of these within-ego relationships, Figure 3 plots the within-ego marginal effects on the probability of protection against tie decay across observed values of closed-form cultural matching, open-ended activity matching, and cultural network opacity, conditional on ego fixed effects (relative to a baseline of zero matches or unknown domains). Across an individual’s personal network, increasing closed-form cultural matching from 0 to 6 domain matches progressively increases protection from tie decay by up to 4.7 %, while increasing open-ended activity matching from 0 to 5 shared activities elevates protection by up to 4.6 %. In contrast, cultural network opacity imposes an escalating penalty on tie durability within personal networks: increasing the number of unknown domains from 0 to 6 reduces protection from tie decay by 8.0 %. This indicates that even after conditioning out time-invariant ego characteristics and unobserved individual traits, cultural matching and opacity continue to predict the decay of social ties.

5 Discussion and Conclusion

5 Discussion

5.1 Summary of Key Findings

~~In this paper, we sought to address a central issue in the study of culture and networks: how do cultural tastes shape the evolutionary dynamics of personal networks?~~

5.1 Summary of Key Results

Using longitudinal ego-network data from the *NetSense* study, ~~we considered~~ ~~I evaluated~~ the empirical implications of a cultural matching model of tie ~~persistence.~~ ~~Our results provide strong evidence that shared cultural tastes are decay in personal networks.~~ ~~The results provide consistent evidence that cultural matching is~~ an important mechanism ~~for stabilizing social ties over time, protecting them from natural decay processes protecting social ties from decay~~ (Burt, 2000). Specifically, both broad, closed-form cultural ~~matches~~ ~~matching~~ and specific, open-ended activity ~~matches~~ ~~matching~~ independently reduce the hazard of tie ~~dissolution~~ ~~decay~~, even after ~~rigorously~~ controlling for dyadic structural embeddedness (triadic closure with other alters). While structural embeddedness itself strongly ~~enhances tie survival~~ ~~protects against tie decay~~ ($OR = 1.12, p < 0.05$), it does not displace

Table 7: Sensitivity Models: Absorbing First Decay and Ego Fixed-Effects

	Absorbing First Decay	Ego FE (clogit)
Closed-Form Matches	1.086* (0.041)	1.046+ (0.027)
Open-Ended Matches	1.038 (0.036)	1.056* (0.025)
Network Opacity	0.963 (0.042)	0.920** (0.029)
Structural Embeddedness (Common Alters)	1.120* (0.055)	1.055 (0.036)
Subjective Closeness: Somewhat	1.254 (0.191)	1.409** (0.155)
Subjective Closeness: Close	2.002*** (0.336)	1.949*** (0.233)
Roommate/Dormmate	0.753 (0.201)	0.810 (0.149)
Friend	0.751 (0.196)	0.806 (0.149)
Frequency: Daily (vs Weekly)	1.154 (0.185)	1.078 (0.096)
Race Homophily: Same Race	1.030 (0.151)	0.926 (0.081)
Tie Duration (Scaled)	0.759 (0.187)	0.820 (0.146)
Tie Duration Sq (Scaled)	1.304 (0.346)	1.202 (0.220)
Num.Obs.	4522	1640
AIC	4209.0	10669.8
BIC	4350.2	10777.8

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Model 1 restricts follow-up strictly to the initial continuous tie spell until first tie decay or censoring, excluding all subsequent recurrent/rekindled spells. Model 2 stratifies the likelihood by ego (conditional logit), isolating within-ego variation. Both models adjust for alter gender and wave transition fixed effects; Model 1 also includes ego gender and an ego random intercept.

~~cultural matching. Furthermore, we~~ the effect of cultural matching; rather, it serves as a complementary protective mechanism.

Furthermore, I identified cultural network opacity—the ~~lack of mutual cultural knowledge as a yet-to-be-considered, but nevertheless independent relational liability~~ ego’s lack of knowledge of the alter’s tastes and practices—as an independent cultural mechanism that accelerates tie decay. Crucially, ~~we found that the protective effect of I~~ found that protection from tie decay via closed-form cultural matching and the ~~detrimental~~ accelerating effect of cultural network opacity are most pronounced among ~~strong ties~~ strong ties as measured by subjective closeness (Marsden and Campbell, 1984). For these ~~subjectively close affectively significant~~ relationships, cultural ~~alignment acts as a vital~~ core for tie survival, whereas weak ties appear so fleeting that their matching serves as an important mechanism protecting ties from decay. Weaker, less affectively significant ties, on the other hand, have such an elevated baseline hazard of decay ~~is high regardless of matching in these domains. In contrast~~ that cultural matching can do little to protect them from decaying. Notably, the protective ~~effects effect~~ of specific, ~~open-ended matches remain~~ relatively open-ended activity matching—getting at more granular and specific types of cultural commonalities—against tie decay remains consistent across both weak and strong ties.

5.2 ~~Theoretical Limitations~~ and ~~Substantive Interpretations~~ Suggestions for Future Work

Despite these promising findings, several limitations point toward fruitful avenues for future research. First, regarding operationalization and measurement constraints, the current measurement approach for closed-form cultural matching aggregates across broad domains of lifestyle and taste, with little variance in their overall legitimacy and consecration. Future research could further disaggregate these domains to examine whether traditionally high-status cultural genres (e.g., classical music, highbrow literature) confer distinct relational advantages compared to more popular or subcultural domains (DiMaggio, 1987; Lizardo, 2006; M... . In addition, while the measures focus primarily on positive cultural matching and lack of knowledge (opacity), future work should systematically investigate the role of shared active cultural *dislikes* and mutual distastes as distinct symbolic boundaries in tie decay (Edelmann and Vaisey, 2014).

Second, the network measures capture ego’s *perceived* relational landscape. While egocentric network data are uniquely suited for measuring an actor’s cognitive orientation, subjective closeness, and perceived cultural knowledge about their alters, they capture only one side of the dyad. In non-reciprocated or unidirectional ties, relational orientations may be asymmetric, reflecting underlying differences in social status, popularity, or aspirational connection (Carley and Krackhardt, 1996). In such cases, an ego may actively work to maintain a culturally matched tie with a higher-status peer even if the alter does not reciprocate the nomination (McFarland et al., 2014). Although ego fixed-effects and dyadic controls account for an ego’s general nomination patterns, future investigations with complete dyadic reciprocity data can formally evaluate how cultural matching interacts with status differentials and reciprocal confirmation.

Third, an important question concerns the *temporal horizon* of cultural matching beyond the college context. The panel follows students through their undergraduate trajectory across semester intervals, capturing the critical period when campus ties are actively formed and pruned. However, a compelling open question is whether cultural matching continues to sustain social ties across major post-college life transitions—such as graduation, geographic dispersal, labor market entry, and family formation. When individuals no longer share the daily scaffolding of an institutional campus environment, does deep cultural matching provide the necessary conversational and ritual currency to protect distant ties from decay across decades? Alternatively, do enduring college ties persist primarily via accumulated relational history and institutional memory, rendering cultural matching less decisive once physical co-presence ends? Future long-term multi-decade panel studies will be vital for determining the life-course boundaries of cultural matching.

Finally, the empirical focus on a college-age cohort during emerging adulthood delineates specific institutional scope conditions for these claims. The transition to college represents a unique developmental and institutional ecology characterized by high baseline network churn, rapid relational renegotiation, and geographic concentration (McFarland et al., 2014). While this setting provides an ideal testing ground for observing tie decay and protection from decay in short- to medium-term time scales within a single ecology, future research should examine whether cultural matching operates similarly to protect ties from decay across different life stages and institutional domains. In particular, comparative studies in workplace organizations (McFarland et al., 2014; Kim et al., 2024), civic associations, or among older adults (Cornwell et al., 2008) could shed light on how the capacity of cultural matching to protect against tie decay varies across differing levels of structural constraint.

5.3 Implications: Cultural Capital and Relational Maintenance

These findings carry significant theoretical implications for ~~how we understand~~ understanding the relationship between tastes and networks, particularly regarding the role of tie strength. ~~Our~~ The results heavily favor the empirical implications of the ~~strong-tie matching prediction~~ (~~Strong-Tie Matching Hypothesis over the 3a~~) over the weak-tie scaffolding prediction (~~Weak-Tie Scaffolding Hypothesis - 3b~~).

Aligning with previous research (Edelmann and Vaisey, 2014; Vaisey and Lizardo, 2010), ~~our longitudinal survival~~ the longitudinal event history framework reveals that ~~deep, institutionalized~~ emerging friendships require ongoing cultural ~~resonance~~ matching to sustain intimacy ~~and prevent tie decay~~. It is the *strong* ties that are highly sensitive to cultural matching and opacity. Weak ties, suffering from a high liability of newness, rapidly decay with or without ~~shared tastes~~ cultural matching. These results extend recent observations in life course and network research ~~demonstrating~~ showing that while significant network turnover is a ubiquitous feature of critical life transitions, individuals actively curate their core networks to maintain stable, supportive ties (Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024). ~~More broadly, our findings challenge the assumption of the pure network-transmission perspective, which treats cultural tastes as merely fluid by-products of existing social ties (Mark, 2003). Instead, our findings align more closely with the cultural capital perspective (Bourdieu, 1984), suggesting that tastes function as durable, portable resources that individuals actively use to select, maintain, and discard social relationships.~~

Returning to the puzzle posed in the introduction—how ~~do we to~~ account for durable cultural profiles in the face of volatile networks (Bourdieu, 1984; Holt, 1998)—~~our—these~~ results help to close the loop. If persons use relatively stable cultural dispositions to selectively maintain their networks, it suggests that cultural capital is not merely a passive status marker or a “thin” effluvium of network position. Rather, it is an active navigational tool for managing social environments. People churn through weak ties at a high baseline rate, but they rely on their durable cultural compass to actively curate and protect the core relationships that matter from decay.

Furthermore, ~~our—these~~ findings speak directly to the classic distinction between choice homophily and induced homophily (McPherson et al., 2001). If cultural matching actively prevents tie decay and opacity accelerates it, then a substantial portion of the observed homophily in established social networks is likely due to selective ~~retention—protection against tie decay~~ (choice homophily) rather than pure structural constraint or network-induced adaptation (Lizardo, 2006; Lewis and Kaufman, 2018). By ~~retaining—protecting~~ culturally matched ties from decay and shedding incompatible ones over time, networks become homophilous ~~through—via~~ an active evolutionary sorting process (Rivera et al., 2010).

~~In this paper, we conceptualize cultural matching as the essential structural core of close relationships.~~ While early network theories emphasized structural constraints or multiplexity (Burt, 2000), ~~our—(Hammer, 1985; Davis, 1963), the~~ findings suggest that ~~shared tastes may be what gives close relationships their staying power~~ cultural matching serves as an additional micro-interactional mechanism that, alongside structural embeddedness, protects close relationships from decay. In the absence of ~~mutual cultural knowledge—the ego’s cultural knowledge of the alter~~ (high opacity), even subjectively close ties face a higher hazard of ~~dissolution—decay~~. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles.

Beyond network theory, these findings carry practical implications for institutions aiming to foster social cohesion. For university administrators and community organizers seeking to reduce social isolation, ~~our—these~~ results underscore that ~~mere proximity or—physical proximity and~~ shared structural affiliations are ~~insufficient—most effective when paired with opportunities for common cultural interests and activities~~ to sustain durable relationships. Facilitating environments where individuals can discover and activate ~~shared cultural interests—from cultural matching—from~~ niche clubs to diverse recreational programs—may be essential for transforming fleeting initial encounters into resilient, long-term support networks.

5.4 ~~Limitations and Future Research~~

~~Despite these promising findings, several limitations point toward fruitful avenues for future research. First, our empirical focus on a collegiate cohort during emerging adulthood delineates specific boundary conditions for our claims. The college transition represents a uniquely dynamic developmental and institutional ecology characterized by high baseline network churn, rapid relational renegotiation, and geographic concentration. While this setting provides an ideal testing ground for observing tie decay and selective retention in real time, future research should examine whether cultural matching operates similarly as a stabilizing force across different life stages and institutional domains. In particular, comparative studies in workplace organizations, civic associations, or among older adults~~

(Cornwell et al., 2008) could illuminate how the protective capacity of cultural matching varies across differing levels of structural constraint.—

Second, an important question concerns the *temporal horizon* of cultural matching beyond the collegiate context.—Our panel follows students through their undergraduate trajectory, capturing the critical period when campus ties are actively formed and pruned. However, a compelling open question is whether cultural matching continues to sustain social ties across major post-collegiate life transitions—such as graduation, geographic dispersal, labor market entry, and family formation.—When individuals no longer share the daily scaffolding of an institutional campus environment, does deep cultural alignment provide the necessary conversational and ritual currency to keep distant ties alive across decades? Alternatively, do enduring collegiate ties persist primarily through accumulated relational history and institutional memory, rendering cultural matching less decisive once physical co-presence ends? Future long-term multi-decade panel studies will be vital for determining the life-course boundaries of cultural matching.—

Third, our network measures capture the *ego-perceived* relational landscape.—While egocentric network data are uniquely suited for measuring an actor’s cognitive orientation, subjective closeness, and perceived cultural knowledge about their alters, they capture only one side of the dyad. In non-reciprocated or unidirectional ties, relational orientations may be asymmetric, reflecting underlying differences in social status, popularity, or aspirational connection. In such cases, an ego may actively work to maintain a culturally matched tie with a higher-status peer even if the alter does not reciprocate the nomination. Although our ego fixed-effects and dyadic controls account for an ego’s general nomination patterns, future socioecentric investigations with complete dyadic reciprocity data can formally evaluate how cultural matching interacts with status differentials and reciprocal confirmation.—

Fourth, the current operationalization aggregates cultural matches across broad domains of lifestyle and taste. Future research could further disaggregate these domains to examine whether high-status cultural genres (e.g., classical music, highbrow literature) confer distinct relational advantages compared to widely popular or subcultural domains (DiMaggio, 1987).—In addition, while our measures focus primarily on positive cultural alignment and lack of knowledge (opacity), future work should systematically investigate the role of shared cultural *dislikes* and mutual distastes as distinct symbolic boundaries in tie dissolution.—

Finally, while our discrete-time event history framework provides a rigorous foundation for modeling the hazard of tie decay net of unobserved heterogeneity, future investigations with continuous digital trace streams would benefit from estimating stochastic actor-oriented models (SAOMs) or continuous-time network dynamics (Hachen et al., 2022). Such models can simultaneously estimate the reciprocal, co-evolutionary processes through which social ties induce taste convergence (influence) and pre-existing cultural dispositions guide tie selection and pruning (selection).—

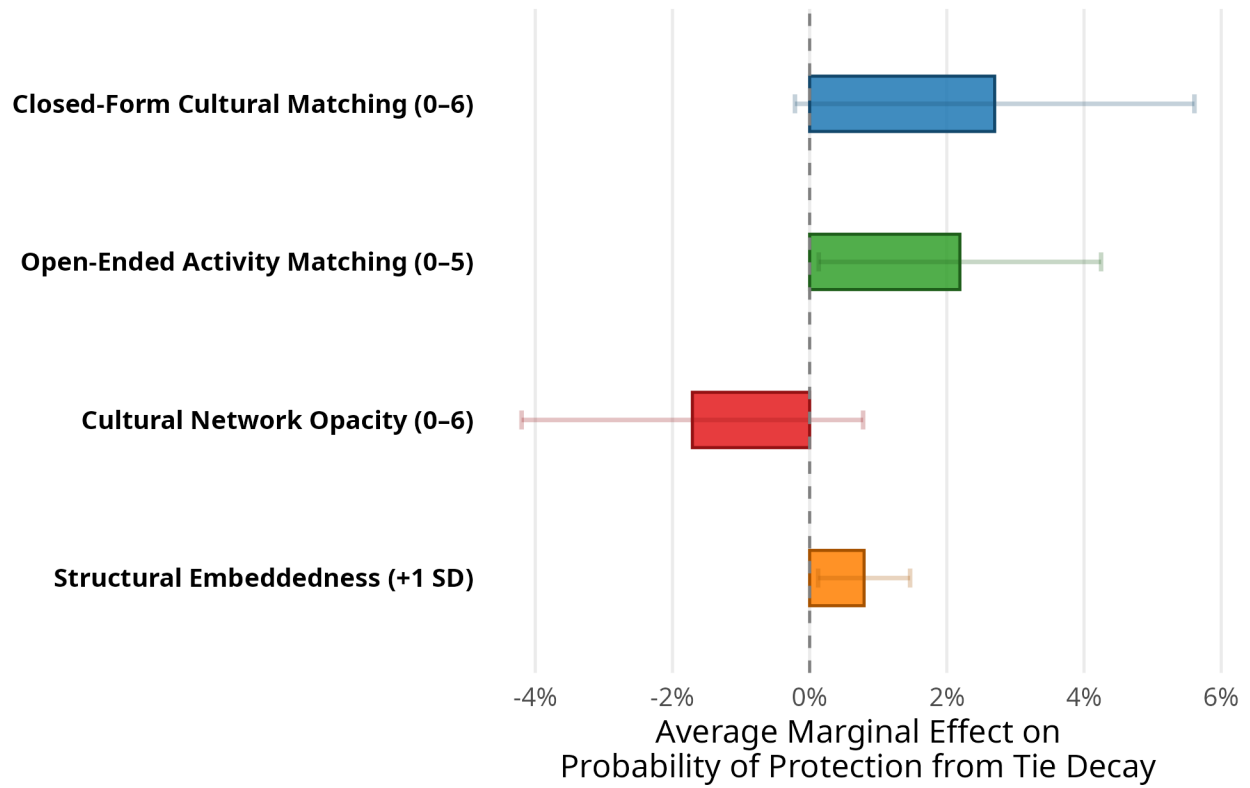


Figure 1: ~~Predicted Probability of Tie Persistence by Cultural Matching and Cultural Network Opacity.~~ Predicted Probability of Protection from Tie Decay by Cultural Matching and Cultural Network Opacity.

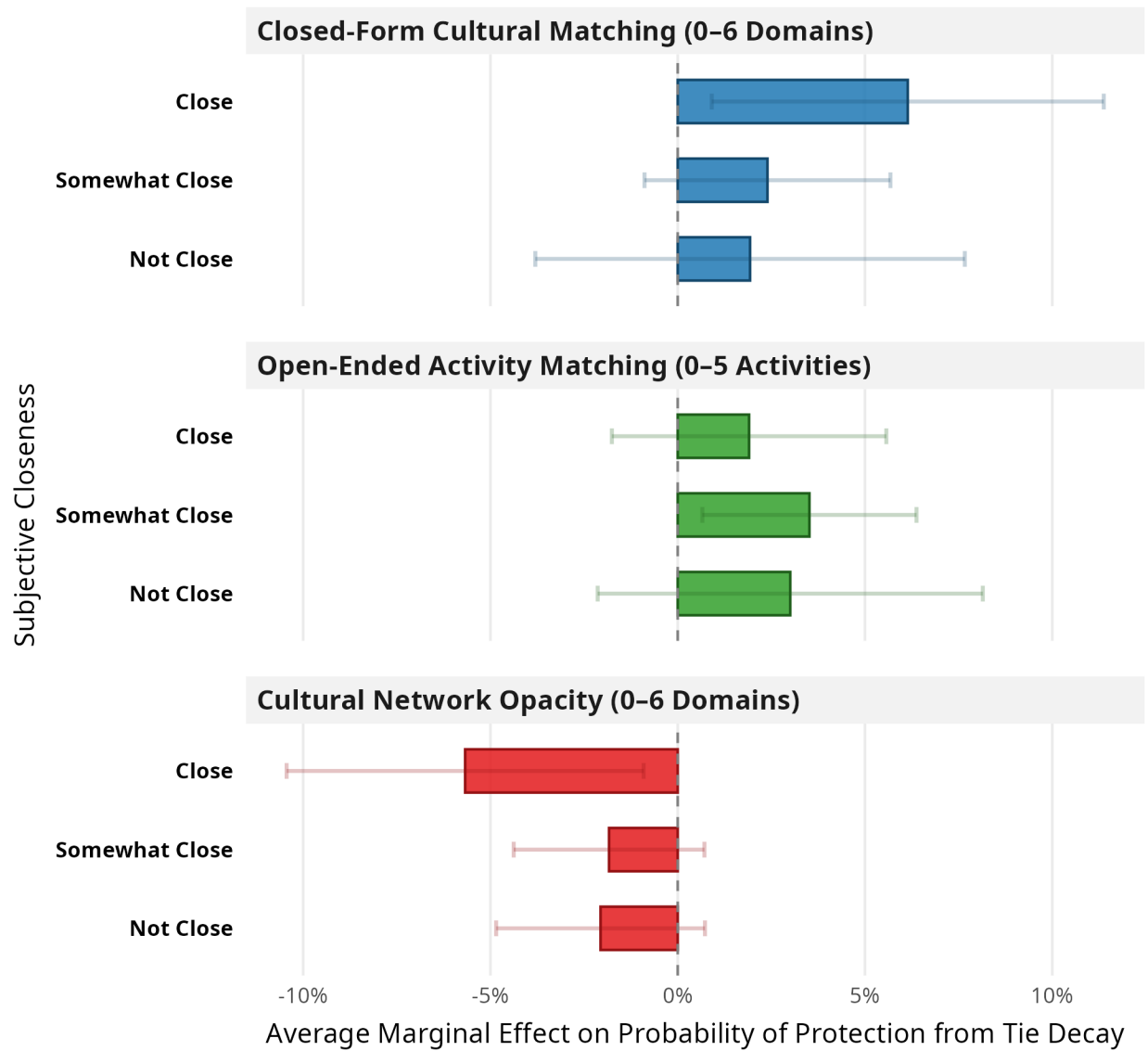


Figure 2: ~~Predicted Probability of Tie Persistence by Cultural Matching, Cultural Network Opacity, and Subjective Closeness.~~ Predicted Probability of Protection from Tie Decay by Cultural Matching, Network Opacity, and Subjective Closeness.

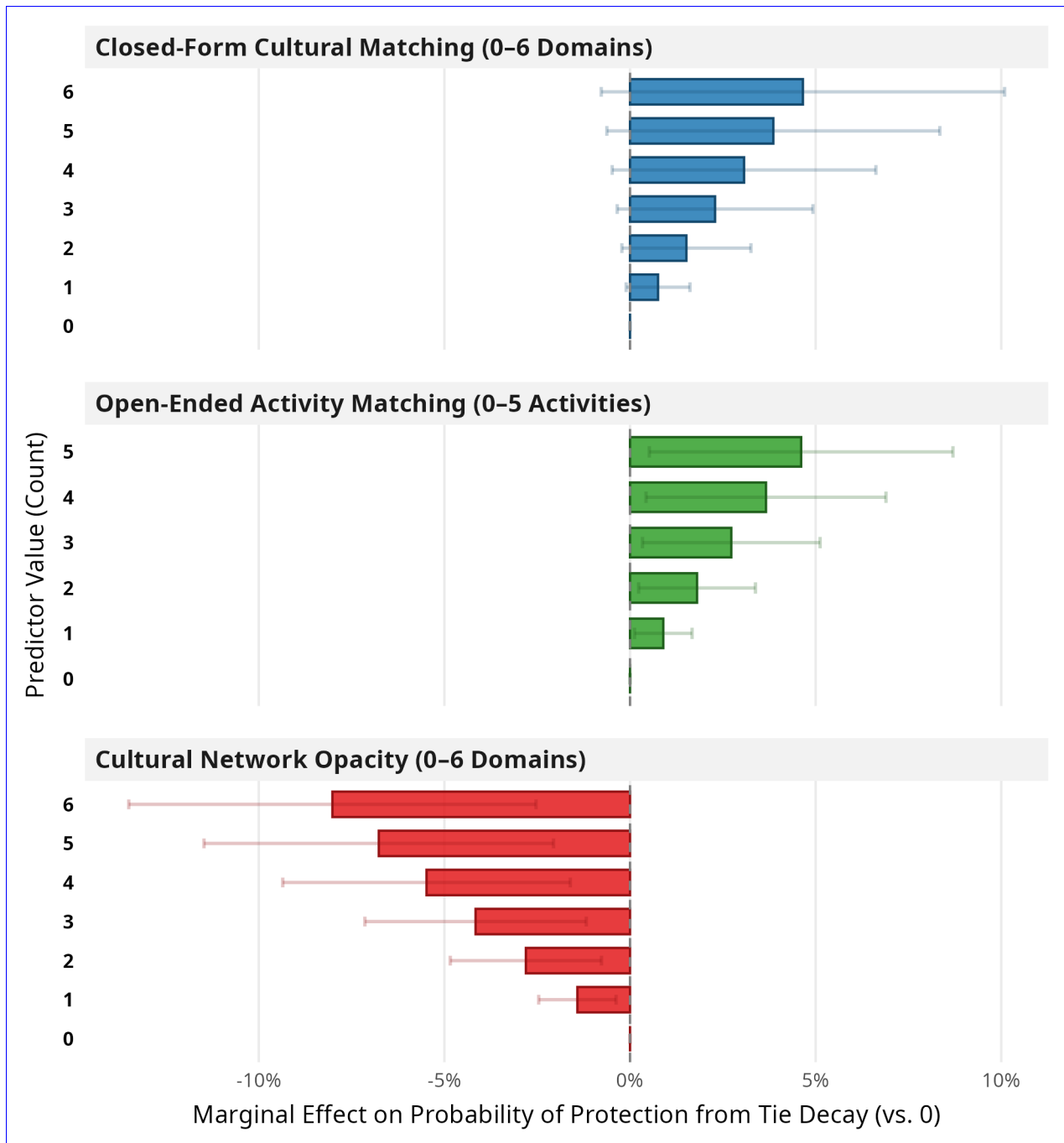


Figure 3: Within-Ego Marginal Effects on Protection from Tie Decay by Cultural Matching and Opacity Counts (Ego Fixed-Effects).

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